



[MACHINE] LEARNING GALAXY EVOLUTION WITH DREAMS



ALEX M. GARCIA
& DREAMS COLLABORATION

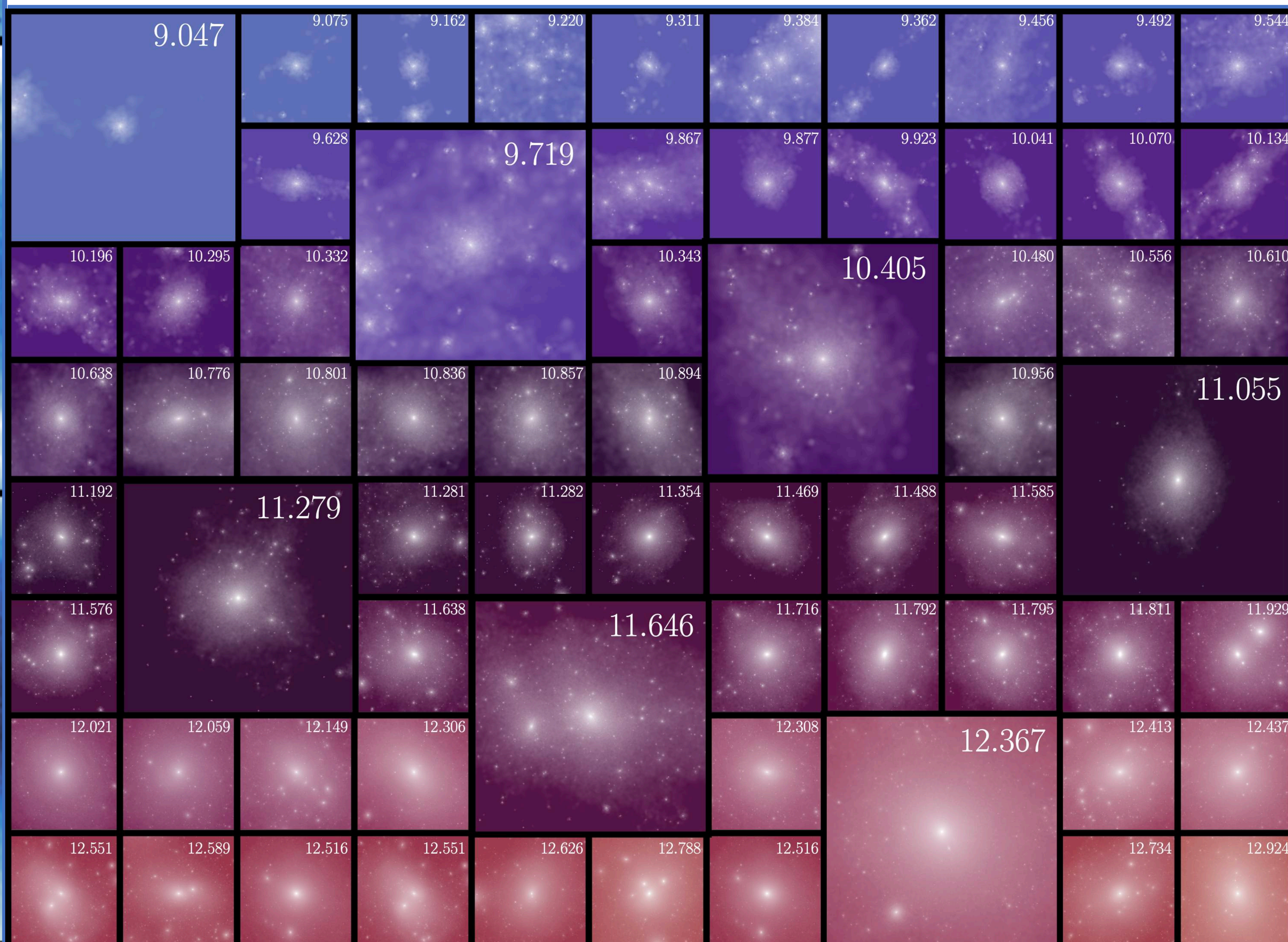
**DARK MATTER AND ASTROPHYSICS WITH
MACHINE LEARNING AND SIMULATIONS**

DARK MATTER

- ▶ INVISIBLE MATTER THAT FORMS THE BACKBONE OF THE UNIVERSE
- ▶ LARGE SYSTEMATIC UNCERTAINTY

ASTROPHYSICS

- ▶ PHYSICS OF STARS, GAS, BLACK HOLES, DUST, ETC.
- ▶ SHAPE THE EVOLUTION OF GALAXIES



THE PROBLEM(S)

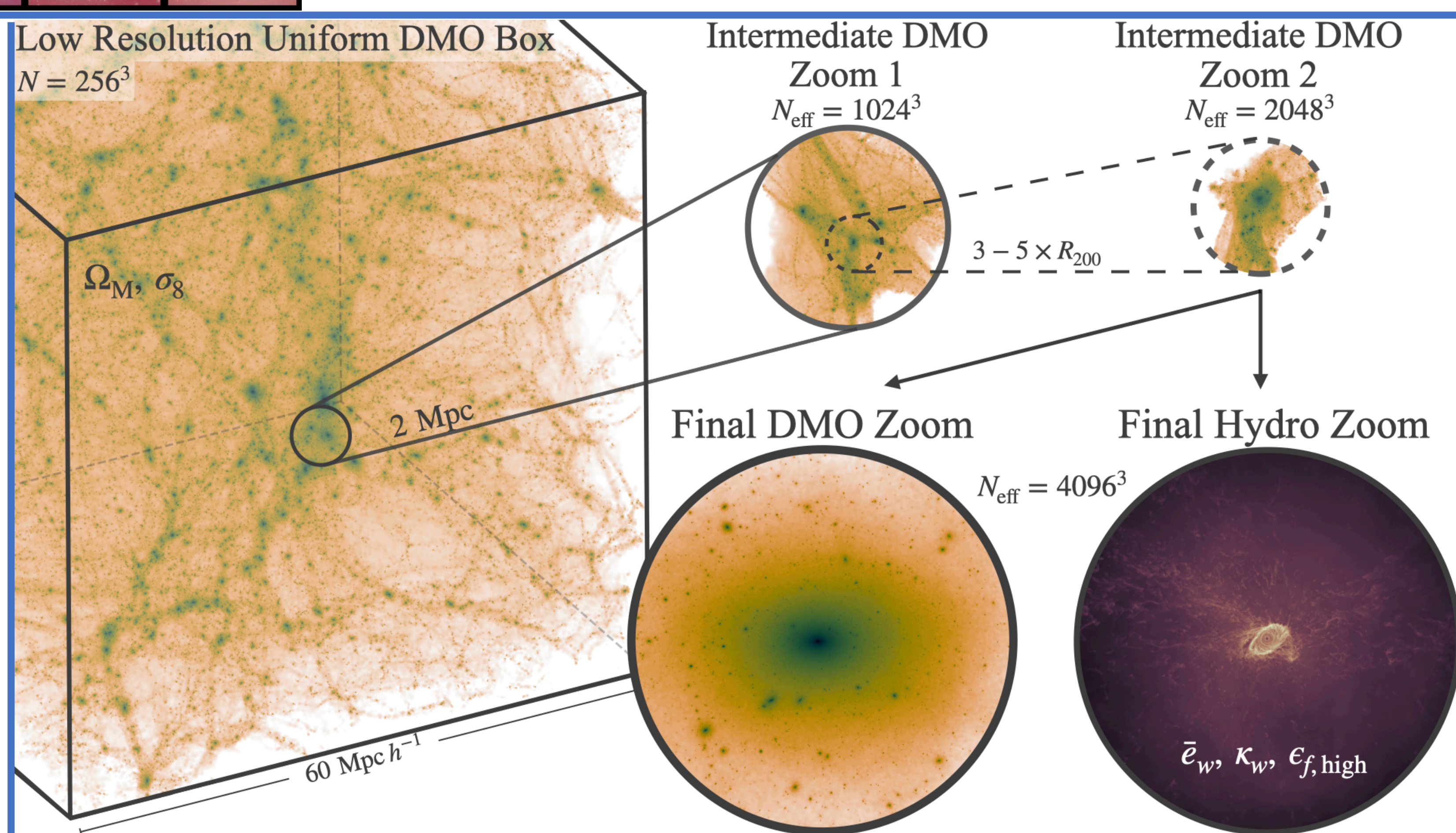
- ▶ WHAT SETS GALAXY PROPERTIES???
- ▶ SIMULATIONS CAN HELP US UNDERSTAND
- ▶ RUNNING MONOLITHIC HIGH RESOLUTION SIMULATIONS IS **EXPENSIVE** !!!
- ▶ RESULTS FROM ONE MODEL MIGHT NOT BE GENERIC!
- ▶ HOW CAN WE STILL LEARN ABOUT THE UNIVERSE???

(LEFT) EXAMPLE HALOS WITH A WIDE RANGE OF MASSES AND ENVIRONMENTS

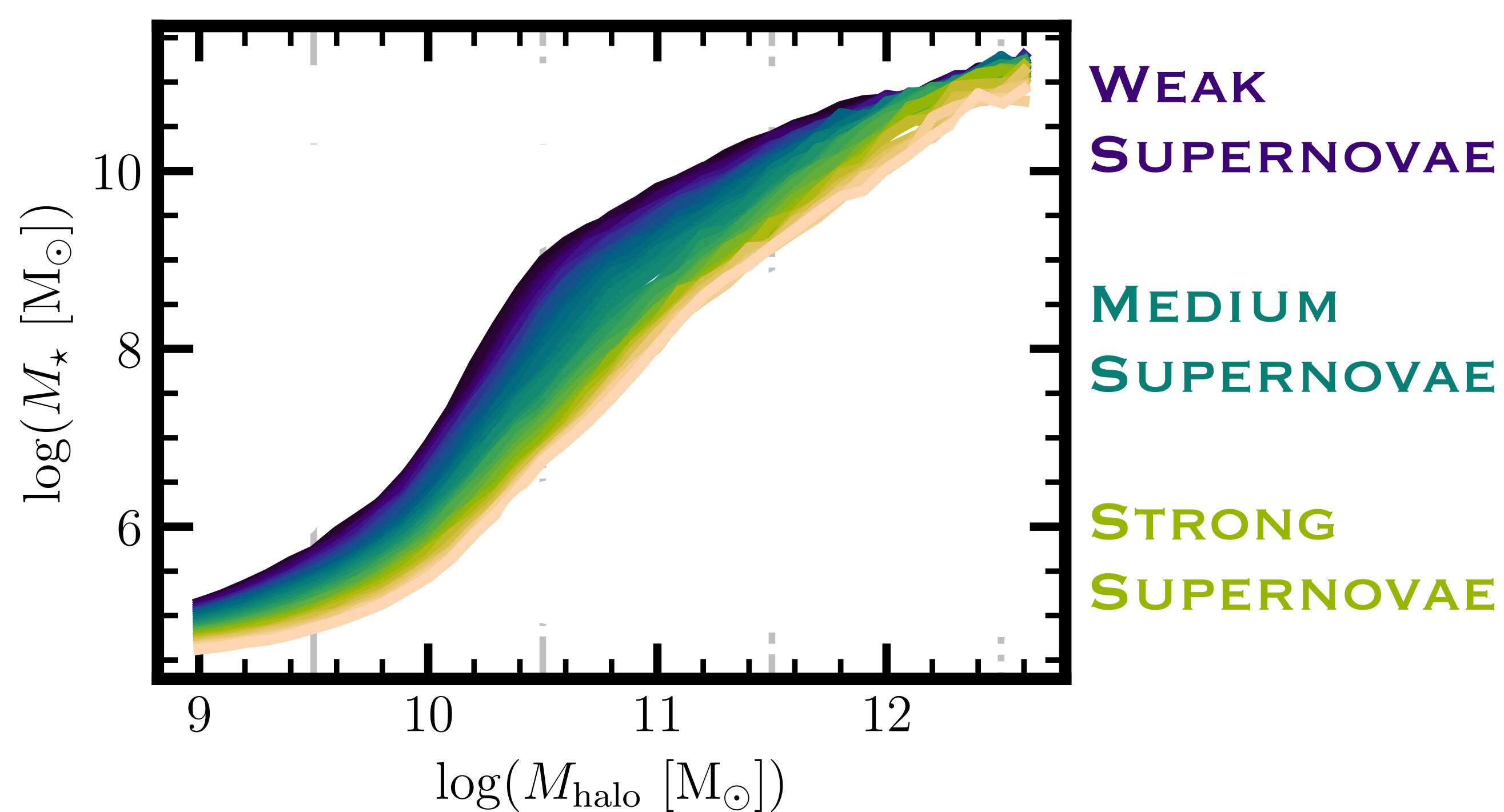
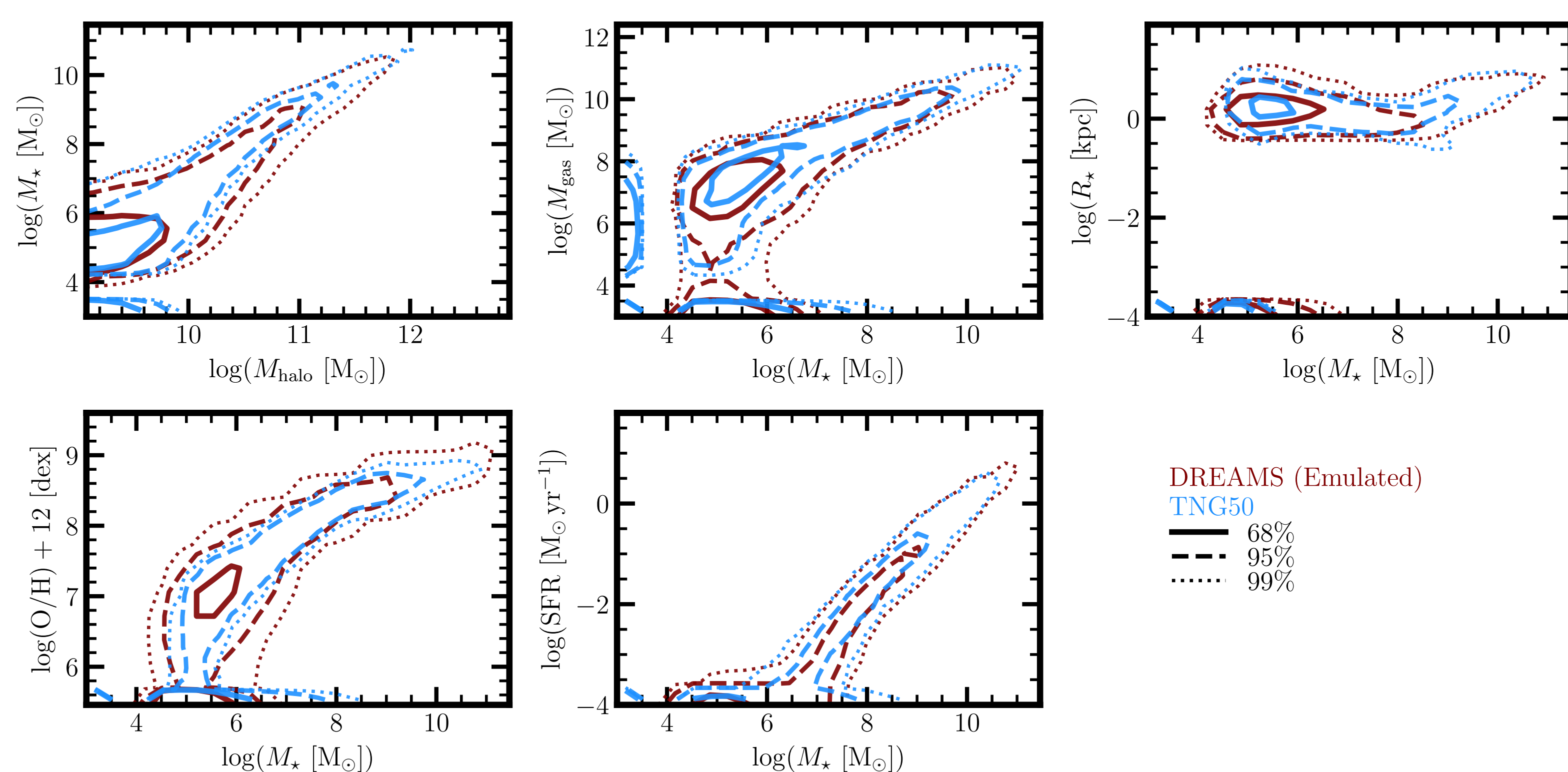
OUR SOLUTION

- ▶ 1000S OF MINI-SIMULATIONS
- ▶ DIFFERENT ASTROPHYSICS AND DARK MATTER IMPLEMENTATIONS
- ▶ MACHINE LEARNING EMULATORS AS HIGH-DIMENSIONAL INTERPOLATORS
- ▶ EFFECTIVE SPEED UP OF **>1000x** !!!

(RIGHT) SELECTION OF MINI-REALIZATION FROM LOW RESOLUTION LARGER VOLUME



(BELOW) SUCCESS OF OUR MACHINE LEARNING MODELS REPRODUCING LARGER VOLUME SIMULATIONS



(ABOVE) EMULATOR ABLE TO MAKE PREDICTIONS FOR SIMULATIONS WE DIDN'T RUN!